

Learning Rate Informed Priors for Neural Network Calibration

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Abstract. Larger learning rates are frequently associated with better generalization, as they encourage exploration of flatter minima in the loss landscape. In this work, we show that the large learning rate phase produces predictive distributions that naturally capture dataset-dependent uncertainty and structured relationships between classes. Across several datasets, we observe that models trained with a large learning rate exhibit significantly lower calibration error early in training, indicating that large learning rate dynamics generate probability distributions that are more faithful to the intrinsic uncertainty of the data labels. Building on this observation, we introduce Learning Rate Informed Priors (LRIP), a simple training strategy that reuses the predictive distributions obtained at the end of the large learning rate phase as improved supervision targets. LRIP replaces uniform or heuristic smoothing with sample-dependent priors extracted directly from the model when its predictions are the most calibrated. Extensive experiments across different modalities and learning-rate schedules demonstrate that LRIP consistently improves calibration while maintaining competitive accuracy. These findings highlight a new perspective on the role of large learning rates: beyond optimization benefits, they induce informative predictive uncertainty that can be directly leveraged to build more reliable and better-calibrated models.

1 Introduction

The learning rate (LR) plays a central role in deep neural network training. Its value affects optimization stability, the scale of parameter updates, and the type of solutions reached at convergence. A large learning rate typically refers to a regime in which the LR is set close to the maximum value that still allows stable training, often several orders of magnitude larger than the values used during the refinement phase. Such regimes have been extensively and experimentally studied, notably in the context of super-convergence [25], where high learning rates induce rapid exploration of the loss landscape and lead to faster or better generalization. Consequently, much of the existing literature has focused on

understanding these optimization dynamics. Existing explanations mainly emphasize optimization aspects, such as escaping sharp minima, increased gradient noise, or transitions in dynamical regimes [11, 15, 10]. While these perspectives describe how parameters evolve, they do not address what kind of information is present in the model’s predictions during this phase.

In supervised learning, the predicted label confidences is as important as the final predicted label. The quality of this distribution is assessed through, the so-called calibration, which measures how well predicted probabilities reflect true likelihoods. Although calibration is increasingly recognized as a key property for reliable deep models, its behavior under different learning-rate regimes has received little attention. In this work, we study the predictive distributions produced during the large-LR phase and observe that they systematically encode useful information about dataset ambiguity. During this phase, the model assigns probabilities not only to the correct class but also to visually or semantically related classes. These early distributions therefore contain richer structure than one-hot labels or uniformly smoothed targets [26]. Based on this observation, we introduce Learning Rate Informed Priors (LRIP). LRIP extracts the model’s predictive distributions at the end of the large LR phase and reuses them as sample-specific priors once the learning rate decreases. This provides supervision that reflects the uncertainty observed during the exploratory phase of training, without modifying the architecture or increasing memory cost.

Our contributions are as follows:

1. **Characterization of the large LR phase:** we show that early predictions obtained under a large learning rate capture dataset-specific ambiguity and exhibit significantly better calibration than predictions produced later in training.
2. **A training strategy:** we propose LRIP, which reuses the predictive distributions at the large LR stage as informative priors for subsequent training, leading to improved supervision compared to one-hot or smoothed labels [26].
3. **Evaluation across modalities:** we demonstrate that LRIP consistently improves calibration on classification and segmentation tasks, and we compare it favorably to recent calibration-aware methods such as MbLS [17], ACLS [22] and CALS [18].

This work highlights that the large learning rate phase is not only an optimization mechanism, but also a source of informative predictive distributions that can be exploited to enhance model calibration.

2 Related work

Large learning rates in Neural Network training

Large learning rates (LRs) have been widely studied for their impact on optimization behaviour and generalization. Several works analyze how large LR values influence the training trajectory and the types of solutions reached by stochastic gradient descent (SGD) [6, 2], including studies on learning dynamics

[14] and learning order effects [15]. Practical schedules such as cyclical LR [24], warm restarts [19], and large batch scaling rules [7] further highlight the importance of the LR regime for effective training. Our work differs in focus: rather than studying optimization geometry or dynamics, we examine the predictive distributions produced during the large LR phase and how they can be reused.

Calibration-oriented training approaches

Model calibration has been addressed through post-hoc techniques like Temperature Scaling [23, 8, 4] and through training-time regularizers such as Label Smoothing [26], Focal Loss calibration [20], and adaptive or class-conditional formulations including MbLS [17], CALS [18], and ACLS [22]. These methods alter the target distributions or gradients to encourage better confidence estimates during training to improve calibration.

Positioning of our contribution

LRIP is complementary to these methods. Instead of designing new smoothing rules or confidence penalties, LRIP reuses the model’s own predictions obtained during the large LR phase as sample-specific priors. To our knowledge, the idea of exploiting early large LR predictions as informative supervisory targets has not been explored in prior calibration or large LR studies.

3 Training loss and calibration background

3.1 Cross-Entropy loss and model calibration

Given an input $\mathbf{x}$ and its corresponding ground-truth label y , a deep neural network (DNN) outputs a logit vector $\mathbf{z} \in \mathbb{R}^C$, where C denotes the number of classes. The predicted probability for class j , also called label confidence, is obtained through the softmax function:

$$p_j = \frac{\exp(z_j)}{\sum_{k=1}^C \exp(z_k)}, \quad (1)$$

Training is typically performed using the cross-entropy (CE) loss:

$$\mathcal{L}_{\text{CE}} = - \sum_{k=1}^C q_k \log p_k, \quad (2)$$

where $\mathbf{q}$ is the target distribution (usually one-hot). The gradient with respect to the logit vector element z_j is:

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_j} = p_j - q_j. \quad (3)$$

The CE loss encourages the model to minimize the discrepancy between the predicted probability distribution $\mathbf{p}$ and the target distribution $\mathbf{q}$, thereby facilitating the learning of discriminative feature representations. Ideally, the probability p_j corresponding to the true class j should approach one, matching the target q_j . To achieve this, the CE loss drives an increase in the corresponding logit vector element z_j (and consequently in the probability p_j).

Model Calibration and Metrics

Model calibration refers to the agreement between a classifier’s predicted confidence and its empirical correctness. A classifier is said to be well-calibrated when predictions made with a given confidence level are correct with approximately the same frequency. A standard quantitative formulation of this notion is provided by the Expected Calibration Error (ECE) [20], which measures the discrepancy between predicted confidence and observed accuracy.

Consider a set of N predictions, each associated with a predicted label $\hat{y}_i$ and an associated confidence $\hat{p}_i \in [0, 1]$. To estimate calibration, predictions are partitioned into M disjoint confidence bins $\{B_m\}_{m=1}^M$, each corresponding to a fixed confidence interval.

For a given bin B_m , the empirical accuracy is defined as

$$A_m = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbb{1}(\hat{y}_i = y_i),$$

while the average confidence is given by

$$C_m = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i,$$

where $\mathbb{1}(\cdot)$ denotes the indicator function.

The Expected Calibration Error is then defined as

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |A_m - C_m|. \quad (4)$$

Lower values of ECE indicate better calibration. Over-confident models correspond to bins where $C_m > A_m$, while under-confident models satisfy $C_m < A_m$. A closely related variant, Adaptive ECE (AECE)[21], follows the same formulation but uses bins constructed to contain approximately equal numbers of samples, providing a complementary view of calibration behaviour.

Illustrative example (medical thresholding). Consider an automated system for detecting malignant lesions in medical images. Clinicians choose to automatically accept predictions with confidence $\hat{p} \geq 0.90$, and send the rest for manual review.

- **If the model is well-calibrated (low ECE/AECE):** Predictions above 0.90 confidence are truly correct about 90% of the time, meaning that among 100 such predictions the model makes about 90 correct decisions and 10 errors, in average.
- **If the model is over-confident (high ECE/AECE):** The model still reports confidence above 0.90, but its actual accuracy in this range drops in fact to only 60%. Thus, for the same 100 high-confidence predictions, it now produces 60 correct decisions and 40 errors, in average.

This example shows that accuracy alone is insufficient in high-stakes settings: it is the calibration and thus the reliability of the confidence scores that determines whether threshold-based decisions are safe or not.

4 Preliminaries: Effect of a Fixed Learning Rate on Calibration

4.1 Setup

Here we examine how a constant learning rate influences the evolution of predicted confidences during training. By using fixed LR instead of schedules, we isolate the effect of LR magnitude on accuracy, uncertainty, and calibration. We study three LR values per dataset, chosen to represent large, intermediate, and small ranges.

Datasets and Metrics : We conduct experiments on CIFAR-10 [12], Tiny-ImageNet [1], CUB-200-2011 [27], and 20 Newsgroups [13]. CIFAR-10 contains 50,000 training and 10,000 test images across 10 classes. Tiny-ImageNet includes 100,000 training images, 10,000 validation images, and 10,000 test images over 200 classes. CUB-200-2011 consists of 11,788 fine-grained bird images across 200 categories. The 20 Newsgroups dataset contains 20,000 text documents from 20 topics, several of which are semantically close. For all datasets, we report Top-1 Accuracy, Expected Calibration Error (ECE), and Adaptive ECE (AECE). These metrics allow us to measure both discriminative performance and confidence reliability throughout training.

Training Protocol : For CIFAR-10 and Tiny-ImageNet, we use a ResNet-50 [9] architecture; for CUB-200-2011, we use ResNet-101[9]. On 20 Newsgroups, we employ a GPCN architecture [16] with pretrained GloVe embeddings, following the standard network choices commonly adopted in the calibration literature [17, 22]. Batch sizes are set to 128 for CIFAR-10, 64 for Tiny-ImageNet, 16 for CUB-200-2011, and 64 for 20 Newsgroups. All models are trained for 50 epochs with a fixed learning rate throughout the entire run. Optimization is performed with SGD (momentum 0.9, weight decay 5×10^{-4}) for vision tasks and Adam ($\beta_1 = 0.9, \beta_2 = 0.999$) for 20 Newsgroups. This setup isolates the effect of learning rate magnitude on training dynamics and predicted confidences, independent of schedules or annealing.

4.2 Results

Table 1 and Fig 1 show that the magnitude of the learning rate directly influences the model’s calibration behaviour during training. Large learning rates tend to produce lower training ECE and AECE values even when accuracy is still increasing. The predicted confidences remain softer and less concentrated for difficult examples, which indicates that the model captures class ambiguity and similarity early in the optimization process. Intermediate learning rates often yield the highest training accuracy but consistently lead to higher calibration error. Their outputs become sharp earlier in training, which contributes to over-confident predictions. Very small learning rates underfit both accuracy and calibration and do not produce meaningful probability structures. This behaviour appears consistently across the four datasets, suggesting that LR magnitude is a central factor in shaping how uncertainty and confidence evolve during training.

Table 1: Effect of a fixed learning rate on training accuracy and calibration. Best values per dataset are **bolded**.

Dataset	LR	ACC train	ECE train	AECE train
CIFAR-10	0.1	85.34	0.70	0.58
	0.01	94.52	0.17	0.12
	0.001	70.92	0.68	0.67
Tiny-ImageNet	0.1	41.55	1.92	1.89
	0.01	74.09	12.56	12.56
	0.001	33.24	6.49	6.49
CUB-200-2011	0.1	20.13	3.32	3.83
	0.01	57.32	11.50	11.50
	0.001	0.87	0.15	0.32
20 Newsgroups	0.01	65.79	1.60	1.60
	0.001	95.65	0.62	0.69
	0.0001	79.25	6.47	6.45

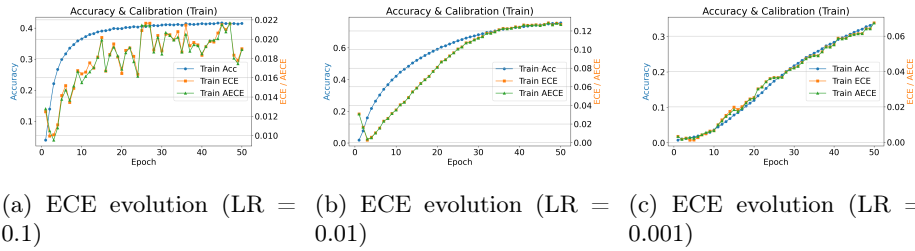


Fig. 1: Effect of a fixed learning rate on Tiny-ImageNet calibration values. Each plot shows how ECE value varies throughout training, highlighting the distinct behaviour induced by LR values in large, intermediate, and small range

4.3 Discussion

These results highlight the distinct role of large learning rates in early learning. Large LRs generate predicted confidences over classes that remain spread out and thus better reflect dataset class ambiguities, leading to substantially lower calibration error during training. Intermediate and small LRs either collapse too quickly into overconfident predictions or fail to explore the data effectively. This observation motivates the next step of our work. If predictions obtained under a large learning rate encode meaningful information about uncertainty and class relationships, then capturing these predictions at key moments during training may provide useful priors for subsequent optimization. The next section details this idea.

5 Proposed Method

The Preliminaries showed that a large learning rate produces predicted confidences that are significantly better calibrated during the early stages of training. This observation drives us to a direct connection with regularization-based calibration methods. Techniques such as Label Smoothing [26] and its adaptive variants attempt to compensate for the limitations of one-hot labels by injecting uncertainty manually. In contrast, the early predictions obtained under a large learning rate arise naturally from the learning capturing relevant a class ambiguities. Our goal in this section is to propose how these predictions can be integrated into the learning objective.

5.1 Label Smoothing as a Probabilistic Objective

Given a one-hot target vector $y \in \mathbb{R}^K$, a prediction vector $p \in \mathbb{R}^K$, and a smoothing factor $\varepsilon \in [0, 1]$, Label Smoothing (LS) [26] replaces the one-hot hard target with a softened distribution. Here, K denotes the number of classes and the index $j \in \{1, \dots, K\}$ refers to a specific class. In LS, the smoothed target is defined as:

$$y_j^{LS} = (1 - \varepsilon) y_j + \frac{\varepsilon}{K}.$$

The corresponding loss is now:

$$\mathcal{L}_{LS} = - \sum_{j=1}^K y_j^{LS} \log p_j,$$

where p_j denotes the predicted class probability defined in Eq. (1).

By expanding this expression and using the definition of the cross-entropy loss in Eq. (2) together with the Kullback–Leibler divergence, the LS objective can be rewritten as:

$$\mathcal{L}_{LS} = (1 - \varepsilon) \mathcal{L}_{CE}(y, p) + \varepsilon D_{\text{KL}}(u_{\text{uni}} \| p) + \varepsilon \log K,$$

where $u_{\text{uni}} = (\frac{1}{K}, \dots, \frac{1}{K})$ is the uniform distribution over the K classes and $D_{\text{KL}}(\cdot \| \cdot)$ denotes the Kullback–Leibler divergence.

This formulation makes explicit that LS combines the standard cross-entropy objective with a KL regularization term toward a fixed prior, yielding a probabilistic interpretation of label smoothing.

5.2 Learning Rate Informed Priors (LRIP)

Results from the above analysis reveal that predicted confidences, when computed with a high learning rate, encode informative signals about dataset class ambiguity. We therefore propose that LRIP replaces the fixed uniform prior used in label smoothing with a data-dependent prior extracted during training in the large LR exploratory phase.

In particular, we propose to link when this prior is collected with a rule only related to the learning-rate schedule. With the proposed rule, LRIP can be applied to any kind of schedule that contains a distinct exploratory phase in which the learning rate is large, either because it increases from a warm-up or because it remains at a plateau followed by a refinement phase where the learning rate decreases. A large LR period appears in standard learning-rate decay schedules as well as in warm-up plus decay schemes, such as the OneCycle policy[25] where the learning rate first increases to a peak before decaying.

In all such schedules, there exists a transition point, denoted t^* , corresponding to the final iteration at which the learning rate attains (or is arbitrarily close to) its maximum value. At this moment, from what was shown in previously, the predicted confidences can be assumed empirically smother and better calibrated. Therefore, the proposed methodology is as follows.

For each training sample x_n , we store the predicted confidences

$$u_{\text{LR}}(x_n) = p(x_n; \theta_{t^*}),$$

where θ_{t^*} denotes the parameters at this large LR boundary. These stored vectors serve as data-dependent priors that reflect the dataset class uncertainties.

During the subsequent refinement phase, training uses the LR-informed loss

$$\mathcal{L}(x_n) = (1 - \varepsilon) \mathcal{L}_{CE}(y_n, p_n) + \varepsilon D_{\text{KL}}(u_{\text{LR}}(x_n) \parallel p_n),$$

preserving the standard discriminative behaviour while encouraging consistency with the informative priors extracted at t^* .

5.3 Training Procedure

Training proceeds in two operational stages (see Fig. 2 for an overview):

Exploratory stage (large learning rate). The model is trained using the standard cross-entropy loss while the learning rate remains at or increases toward its maximum value. This phase—which depends only on the schedule—continues until the learning rate reaches its peak. At this point, indexed by t^* , we record the extracted predicted confidence $u_{\text{LR}}(x_n)$ for every training sample. The proposed rule requires no additional hyperparameters beyond the smoothing strength ε used in the final objective.

Refinement stage (post-peak). When the learning rate begins to decrease, training switches to the LRIP objective. The KL term regularizes the model toward the stored priors, allowing the network to refine decision boundaries while being aware of uncertainties obtained during the first phase exploration.

6 Experiments

In this section, we investigate the effect of reusing the predicted confidences obtained during the large learning-rate phase. Unless otherwise stated, all datasets

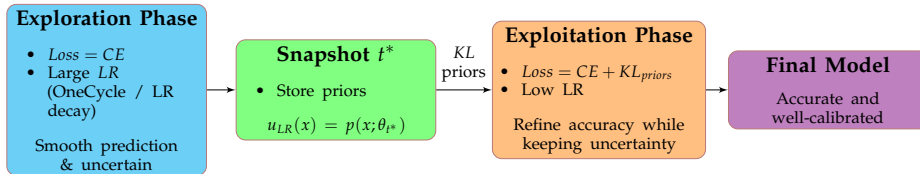


Fig. 2: Overview of the LRIP method. Early predictions obtained at the large learning rate phase are stored as sample-specific priors and reused during the refinement phase through a KL regularization term.

use the same batch sizes and optimizers as specified in Section 4, and the smoothing strength is fixed to $\varepsilon = 0.1$ for all experiments, except for 20 Newsgroups, where $\varepsilon = 0.3$. For image classification and text classification tasks, each model is trained using three different random seeds, and all reported results correspond to the average over these runs. For semantic segmentation on PASCAL VOC 2012 [5], we directly reuse the results reported by MbLS [17] and ACLS [22], obtained under their original experimental protocol and settings.

For each dataset, the snapshot epoch t^* corresponds to the end of the exploratory large learning-rate phase, as determined by the learning-rate schedule. For CIFAR-10, models are trained for 150 epochs using step learning-rate decay at epochs 50 and 100. The first decay marks the transition from exploration to refinement, and we therefore set $t^* = 50$. For Tiny-ImageNet, training follows a 100-epoch schedule with learning-rate drops at epochs 40 and 60, yielding $t^* = 40$. For CUB-200-2011, we adopt a OneCycle learning-rate schedule over 160 epochs with a maximum learning rate of 0.1. Following the standard 30% exploration and 70% refinement split, the peak learning rate is reached at epoch $t^* = 0.3 \times 160 = 48$, where the predictive distributions are stored.

For 20 Newsgroups, the model is trained for 30 epochs using a one-cycle learning rate scheduler with `max_lr = 0.01`, `pct_start = 0.3`, `anneal_strategy = cos`, and `cycle_momentum = True`. The end of the exploratory phase is therefore at $t^* = 9$. For PASCAL VOC 2012, we follow the experimental protocol of MbLS [17]. We use DeepLabV3 [3] with an ImageNet-pretrained ResNet-50 backbone and optimize the model using SGD for 100 epochs with batch size 8, learning rate 10^{-2} , momentum 0.9, and weight decay 5×10^{-4} . The learning rate is divided by 10 at epochs 40 and 80. Since the large learning-rate phase ends at the first decay, we set $t^* = 40$.

Following standard evaluation protocols in the calibration literature [17, 22], we report Expected Calibration Error (ECE, %), Adaptive Expected Calibration Error (AECE, %), and top-1 accuracy (ACC, %) for image and text classification tasks. For semantic segmentation, we report ECE (%), AECE (%), and mean Intersection-over-Union (mIoU, %).

6.1 Results and Analysis

Across all datasets, LRIP consistently improves calibration compared to cross-entropy and classical Label Smoothing, while maintaining competitive accuracy. The gains are visible in image classification, text classification, and semantic segmentation, demonstrating that the class uncertainties captured during the large learning-rate phase generalizes reliably across modalities.

Image Classification: Table 2 summarizes the results on CIFAR-10, Tiny-ImageNet, and CUB-200-2011 using ResNet-50 (R50) and ResNet-101 (R101).

CIFAR-10. LRIP provides the lowest calibration errors among all methods. For ResNet-50, LRIP achieves an ECE of **1.25** compared to 3.35 (CE), 3.66 (LS), and 1.44 (MbLS). For ResNet-101, LRIP again obtains the best ECE of **1.22**, outperforming CE (3.37), LS (3.43), and MbLS (1.58). Accuracy remains comparable to all baselines, indicating that the calibration improvements do not come at the expense of discriminative performance.

Tiny-ImageNet. This dataset contains more fine-grained ambiguity, yet LRIP continues to provide competitive calibration. With ResNet-50, LRIP achieves ECE = **1.55**, second only to ACLS (1.35) and close to MbLS (2.19). With ResNet-101, LRIP reaches ECE = 1.53, again close to the best-performing ACLS (1.21) and significantly better than CE (4.97) or LS (2.14). These results show that priors extracted under large learning rates remain informative even in high-entropy settings with many classes.

CUB-200-2011. CUB is a fine-grained dataset with strong inter-class similarity. LRIP achieves the best calibration among all methods, with ECE = **1.77**, outperforming CE (9.53), LS (3.89), MbLS (2.10), and ACLS (2.81). Although accuracy is slightly lower for LRIP, the marked improvement in calibration indicates that the large LR predictions capture meaningful class ambiguities in this challenging dataset. CALS [18] is omitted for CUB-200-2011 because it requires a validation set to compute class-adaptive margins, while the dataset is commonly used with only training and test partitions.

Text Classification: Table 3 reports results on 20 Newsgroups. LRIP clearly outperforms all competing approaches in calibration, reducing AECE to **2.99**, compared to 16.32 (CE), 4.75 (LS), 4.83 (MbLS), and 8.16 (ACLS). Accuracy remains effectively identical across all methods, demonstrating that the calibration improvement is not due to changes in predictive performances but to a better uncertainty napshot from the large LR phase.

Image Segmentation: On PASCAL VOC 2012 (Table 4), LRIP again improves calibration. It reduces ECE from 8.26 (CE), 9.35 (LS), 7.94 (MbLS), and 7.29 (ACLS) down to **6.59**. AECE shows the same trend, decreasing to **6.70**. Importantly, LRIP preserves segmentation performance, achieving an mIoU (71.17) comparable to all baseline methods, confirming that the calibration gains do not come at the expense of segmentation accuracy.

Table 2: Image tasks: Results on CIFAR-10, Tiny-ImageNet and CUB-200-2011. We report top-1 accuracy (ACC), expected calibration error (ECE), and adaptive ECE (AECE).

Method	CIFAR-10 (R50)			CIFAR-10 (R101)			Tiny-ImageNet (R50)			Tiny-ImageNet (R101)			CUB-200-2011 (R101)		
	ACC $\uparrow$	ECE $\downarrow$	AECE $\downarrow$	ACC $\uparrow$	ECE $\downarrow$	AECE $\downarrow$	ACC $\uparrow$	ECE $\downarrow$	AECE $\downarrow$	ACC $\uparrow$	ECE $\downarrow$	AECE $\downarrow$	ACC $\uparrow$	ECE $\downarrow$	AECE $\downarrow$
CE	94.65	3.35	3.35	94.73	3.37	3.36	64.77	3.37	3.35	65.62	4.97	4.36	68.97	9.53	9.53
LS [26]	94.66	3.66	4.00	94.76	3.43	4.12	64.81	2.22	2.27	65.97	2.14	2.18	70.22	3.89	3.90
MbLS [17]	94.45	<u>1.44</u>	3.42	94.75	<u>1.58</u>	3.36	64.69	2.19	2.30	65.69	<u>1.39</u>	<u>1.18</u>	69.72	<u>2.10</u>	<u>2.14</u>
ACLS [22]	94.68	2.26	<u>3.02</u>	94.57	2.53	<u>3.31</u>	64.83	1.35	1.31	65.72	1.21	1.04	69.84	2.81	2.64
CALS [18]	94.17	2.34	3.10	94.20	2.65	3.23	65.12	4.11	3.98	66.18	3.91	3.79	-	-	-
LRIP	94.24	1.25	1.21	94.51	1.22	1.24	64.30	<u>1.55</u>	<u>1.55</u>	65.25	1.53	1.42	67.27	1.77	2.00

Table 3: Results on the testing set of the 20 Newsgroups dataset.

	CE	LS [26]	MbLS [17]	ACLS [22]	LRIP
ACC $\uparrow$	69.61	70.69	71.28	71.31	70.67
AECE $\downarrow$	16.32	4.75	4.83	8.16	2.99

Summary: Across all datasets and architectures, LRIP consistently yields lower ECE and AECE than CE and LS, and often surpasses more complex adaptive smoothing methods such as MbLS or ACLS. Unlike LRIP, MbLS and ACLS rely on margin-based formulations, where calibration is enforced through explicit class-dependent margins. These margins are sensitive to dataset characteristics and model architecture, which can limit their transferability across tasks. By contrast, LRIP avoids explicit margin design and maintains stable calibration improvements across vision and language benchmarks, highlighting its robustness and general applicability.

6.2 Discussion

Reliability diagrams

The reliability diagrams in Fig. 3 illustrate clear differences in calibration behaviour across methods. On both CIFAR-10 (top row) and CUB-200-2011 (bottom row), cross-entropy (CE) exhibits strong over-confidence: the predicted confidence consistently exceeds the empirical accuracy, resulting in large gaps between the blue (expected) and red (output) bars. Label Smoothing reduces this effect but remains imperfect, particularly in low- and mid-confidence bins where the model systematically overestimates its correctness.

MbLS and ACLS partially correct these deviations but their behaviour is irregular: several confidence bins show abrupt fluctuations, and both methods remain over-confident in the high-confidence region. This reflects the sensitivity of adaptive margin-based regularizers to class imbalance and dataset granularity.

Table 4: Results on PASCAL VOC 2012 segmentation benchmark. We report mean IoU (mIoU), expected calibration error (ECE), and adaptive ECE (AECE).

	CE	LS [26]	MbLS [17]	ACLS [22]	LRIP
mIoU $\uparrow$	70.92	71.00	71.20	70.63	71.17
ECE $\downarrow$	8.26	9.35	7.94	7.29	6.59
AECE $\downarrow$	8.23	9.95	7.99	7.28	6.70

In contrast, LRIP produces diagrams that remain close to the ideal diagonal. The accuracy and confidence curves align across all bins, with significantly smaller gaps than any competing method. This indicates that the priors extracted during the large LR phase impose a coherent probabilistic structure that persists throughout training.

The effect is consistent across the two datasets: even on CUB-200-2011, where fine-grained classes induce high uncertainty, LRIP maintains nearly monotonic and well-aligned reliability curves.

Overall, the diagrams confirm that LRIP achieves stable and globally calibrated predictions, whereas other methods either reduce over-confidence only partially (LS, ACLS) or introduce irregular calibration patterns (MbLS). The smoothness and closeness to the diagonal highlight that early large LR phase provides a reliable source about class uncertainty for improved regularizing during later training.

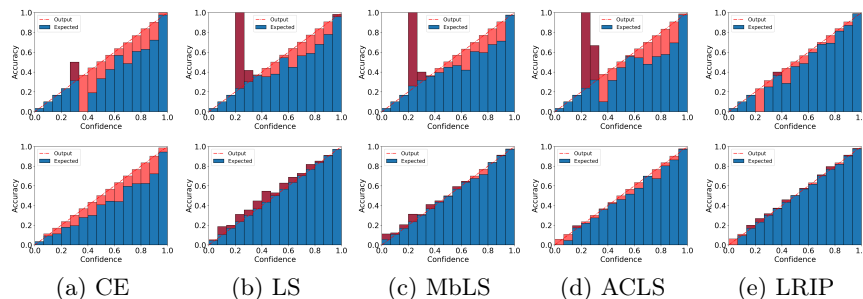


Fig. 3: Comparisons of reliability diagrams for cifar10 (top) and CUB-200-2011(bottom).

Ablation Study: Importance of the large LR Snapshot

To evaluate whether the effectiveness of LRIP truly depends on the predictive distributions generated during the large learning rate phase, we perform an ablation study in which the priors are extracted not at t^* but at the second learning-rate decay. This delayed variant, denoted **SD** in the following, captures

predictions at epoch 100 for CIFAR-10 and at epoch 60 for Tiny-ImageNet, where the learning rate is already small and the model has become substantially more confident.

Table 5 summarizes the results. On CIFAR-10, LRIP achieves an ECE of **1.25**, whereas SD more than doubles it to **2.67**. A similar trend appears in AECE (1.21 vs. 2.63). Accuracy is slightly higher under SD (94.55 vs. 94.24), confirming that the deterioration is not due to discriminative performance but at the cost of a loss of relevance regarding predicted confidences.

On Tiny-ImageNet, LRIP again provides better calibration, with ECE = **1.55** compared to **2.97** for SD. AECE follows the same pattern (1.55 vs. 2.86). Here too, accuracy slightly increases under SD, indicating that late-stage predictions are sharper but less reliable.

These results confirm that the priors extracted during the large learning rate phase carry the dataset-dependent class uncertainties that the proposed LRIP rule is able to better snapshot. In contrast, the predictions obtained after the training enters the low-LR refinement phase collapse towards low-entropy probability distributions, losing the calibration quality observed earlier. This demonstrates that the timing of the snapshot is essential: only the large LR phase seems to provide reliable uncertainty estimates which are needed for effective calibration regularization.

Table 5: Ablation Study: LRIP vs. SD (Second-Decay snapshot). Using late-stage priors significantly worsens calibration.

	CIFAR-10 (R50)		Tiny-ImageNet (R50)	
	LRIP	SD	LRIP	SD
ACC $\uparrow$	94.24	94.55	64.30	65.01
ECE $\downarrow$	1.25	2.67	1.55	2.97
AECE $\downarrow$	1.21	2.63	1.55	2.86

7 Conclusion

In this work, we showed that the predicted confidences obtained during the large learning rate phase contain rich and reliable information about dataset class uncertainty. By capturing these early predictions and using them as data-dependent priors, LRIP provides a simple yet principled alternative to classical calibration techniques based on fixed or hand-crafted priors. Across image classification, text classification, and semantic segmentation, LRIP consistently improves calibration without sacrificing accuracy and without introducing additional hyperparameters beyond the smoothing strength also used in Label Smoothing.

Beyond its empirical performance, this work highlights a broader and largely unexplored perspective: large learning rates are not merely an optimization tool

but also a source of meaningful information about class uncertainties. Leveraging this information opens a new direction for calibration research, in which uncertainties are extracted during training itself rather than imposed externally. An important direction for future work is to investigate the robustness of LRIP under distributional shifts, including out-of-distribution and domain-shift scenarios. Since LRIP relies on uncertainty patterns learned during early training rather than fixed priors or explicit margins, it may offer improved resilience when the test distribution deviates from the training data. Exploring this behavior in safety-critical and real-world deployment settings remains a promising avenue for further research, in our opinion.

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